

Lightning Finals

MNYMO Girls 2025

August 24th, 2025

Rules

- The top individuals will compete two-at-a-time in a single-elimination-style bracket tournament.
- For each match, a problem will be read and projected and the students will have 60 seconds to solve the problem. When a student has an answer ready, they may buzz in and answer and time is paused.

Rules (cont.)

- If it is correct, they gain a point. If it is incorrect, time is resumed and the other student will have the remainder of the time to buzz in and answer.
- The first student to two points wins the match, and this process is repeated until a first-place winner is decided.

Rules (cont.)

- If there is no winner after three problems, the student with the higher score wins. Otherwise, the match will enter sudden death, where the next correct answer wins.

What is $999 \cdot 222$?

Answer: 221778

How many distinct non-degenerate triangles with positive integer side lengths have a perimeter of 9?

Answer: 3

Two fair dice with faces labeled 1 through 6 are rolled. What is the probability that the sum of the numbers on the top faces is 10?

Answer: $\frac{1}{12}$

Rhombus $ABCD$ has diagonals of lengths 10 and 24. What is AB ?

Answer: 13

The number 2^{24} can be written as a^a , where a is a positive integer. Find a .

Answer: 8

Kaining is shopping for school supplies at Target. Folders come in 5 colors, binders come in 3 colors, and notebooks come in 6 colors.

Kaining needs 2 folders, 1 binder, and 5 notebooks, and she will not get the same color for identical items. How many different combinations of school supplies can she buy?

Answer: 180

Alice's desk is shaped like a triangle with side lengths 15, 15, and 18 inches. What is the area of Alice's desk (in square inches)?

Answer: 108

Rectangle $ABCD$ has an area of 30 and satisfies $AC = BD = \sqrt{61}$. Compute the perimeter of $ABCD$.

Answer: 22

Compute the value of

$$100^2 - 99^2 + 98^2 - 97^2 + \dots + 4^2 - 3^2 + 2^2 - 1^2.$$

Answer: 5050

A dartboard is made up of two concentric circles. The inner circle has a radius of 3, and the outer circle has a radius of 5. What is the probability that a dart that lands on the board lands in the inner circle?

Answer: 9/25

Compute the last digit of 824^{2025} .

Answer: 4

Compute $\frac{0.242424\ldots}{0.161616\ldots}$.

Answer: $\frac{3}{2}$

$\triangle ABC \sim \triangle DEF$ with $\frac{AB}{DE} = \frac{7}{9}$. Given that $BC = 21$, what is the length of EF ?

Answer: 27

Given $f(x) = 7x^2 - 22x + 3$, what is the sum of the zeros of $f(x)$? Express your answer as a common fraction in simplest form.

Answer: $\frac{22}{7}$

Sneha's Sandwich Shop is open from 6:00 AM to 7:00 PM every day. On Monday, they serve 84 customers. What is the mean number of customers served per hour? Round your answer to the nearest whole number.

Answer: 6

Michael's Meme Magazine sells magazines at a rate of 1 magazine per day. Each magazine costs \$4 dollars to print and sells for \$34. Given that every magazine is sold, how much net revenue will Michael's Meme Magazine make in August 2025?

Answer: 930

Angie is making a sandwich with 3 options for bread, 4 options for meat, and 5 options for cheese. How many sandwiches can she make?

Answer: 60

How many possible strings of letters can be made from the word *MNYMO*?

Answer: 60

How many positive integers less than 50 are divisible by at least one of 2, 3, or 5?

Answer: 35

How many factors does 2025 have?

Answer: 15

Compute the first digit of 2025^{2025} when it is written in base 2.

Answer: 1

Alice bought 3 pens and 6 pencils for \$0.45.
Angie bought 6 pens and 3 pencils for \$4.50.
How much did Michael spend on 1 pen and 1 pencil?

Answer: \$0.55

Alice, Angie, Elizabeth, Kaining, Michael, and Sneha take 1, 2, 3, 4, 5, and 6 minutes, respectively, to finish a tub of ice cream.

Assuming they all start at the same time, what is the minimum number of minutes such that at least four of them finish a tub at the same time?

Answer: 6

How many two-digit primes have a units digit of 3?

Answer: 6

Sneha lost her precious stash of cookies, each of which contain a distinct, positive integer amount of chocolate chips. Given that the total number of chocolate chips is 30, what is the maximum number of cookies Sneha could have in her stash?

Answer: 7

Angie drove away from MNYMO at 35 mph.
However, she forget her signed copy of EGMO.
Sneha drove after Angie at 45 mph, 1 hour later.
How many hours did it take for Sneha to catch
up to Angie?

Answer: 3.5

Angie has 2000 tabs open on her computer.
Given that she closes one tab every 3 seconds
after 6 PM, what time will it be when there are
no more tabs open?

Answer: 7 : 40 PM

An unfair coin lands heads with probability $x \in (0, 1)$. Given that the probability of flipping the coin tails two times in a row is 0.49, find x .

Answer: 0.3

Sneha is traveling on the coordinate grid and can only move up or right. Starting at the point $(0, 0)$, how many ways are there for her to reach the point $(2, 3)$?

Answer: 10

What is the sum of all possible remainders for
when a perfect square is divided by 5?

Answer: 5

What is the expected sum of the values of the top faces after rolling a fair die (with faces labeled 1-6) 10 times?

Answer: 35

What is the side length of a rhombus with diagonals of length 36 and 48?

Answer: 30